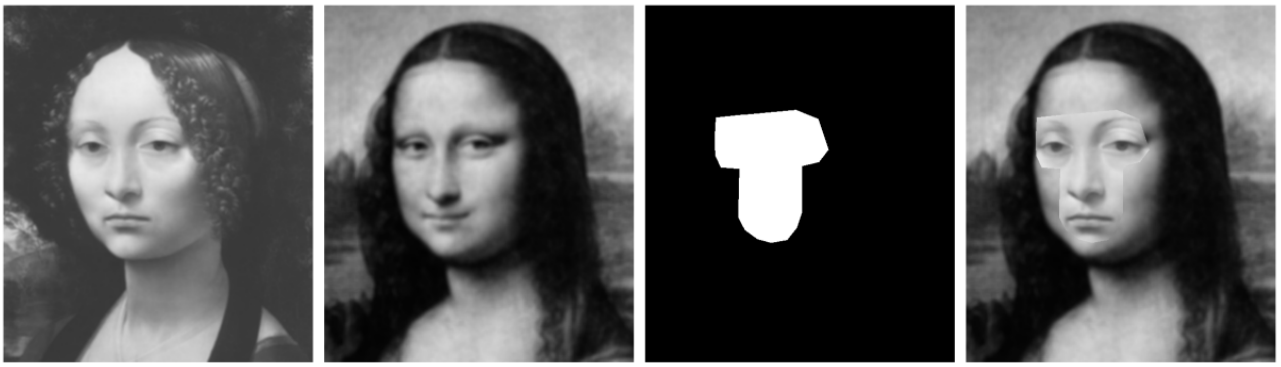
Poisson image editing



An image editing problem. We would like to swap the faces of Ginevra de' Benci (left) and Lisa Gherardini. The image on the right shows the result of composing images by editing the gray levels using the binary mask m (3rd image from the left). The result shows a discontinuity of the gray levels along the boundary of the mask. In the next cells we're going to solve this problem using a technique called *Poisson editing*.

A quadratic energy for image editing

As an application of the conjugate gradient method for minimizing quadratic energies, we will present a method for image edition called Poisson editing. Poisson editing is based on editing the gradient of an image instead of working directly with the gray levels (or with the colors). As we will see, this simplifies many things for certain editing tasks.

We will focus on the following problem: we have two famous portraits by Leonardo Da Vinci, and we would like to interchange the faces of Ginevra de' Benci and Lisa Gherardini. Both portraits have been aligned and scaled, so that the faces are the same size and are in the same position of the image. There is also a mask, which selects the part of the image we want to interchange.

Let $u_1, u_2 : \Omega \to \mathbb{R}$ and $m : \Omega \to \{0, 1\}$ be both images, and the binary mask. Ω is a discrete rectangular 2D grid, the positions of the image pixels: $\Omega = \{1, \dots, M\} \times \{1, \dots, N\}$ (M columns and N rows). Say we want the face of u_1 in u_2. Let us call u the resulting image.

Editing directly the gray levels

The simplest thing to try is to define u as follows:

$$u_{ij} = m_{ij} u_{1,ij} + (1 - m_{ij}) u_{2,ij}, \quad \text{for} \quad (i, j) \in \Omega.$$

This gives the result shown in the figure above. Notice that the editing region is clearly visible! The reason is that one of the images is darker than the other, and this creates a discontinuity with the shape of the mask.

Editing the gradients

A better alternative is to perform the editing by copying *diferences of gray levels* instead of gray levels. Thus, first we will compute the *finite diffence gradient* of both images,

$$\nabla^+ u_{1,ij} = \begin{bmatrix} \nabla_i^+ u_{1,ij} \\ \nabla_j^+ u_{1,ij} \end{bmatrix} \qquad \text{and} \qquad \nabla^+ u_{2,ij} = \begin{bmatrix} \nabla_i^+ u_{2,ij} \\ \nabla_j^+ u_{2,ij} \end{bmatrix}$$

for $i = 1, 2, \dots M, j = 1, 2, \dots N$.

Here ∇_i^+ and ∇_j^+ refer to the forward differences partial derivatives in the direction of i (rows) and j (columns). These are defined as follows:

$$\nabla_i^+ u_{ij} = \begin{cases} u_{i+1,j} - u_{i,j} & \text{if } i < M \\ 0 & \text{if } i = M \end{cases} \qquad \nabla_j^+ u_{ij} = \begin{cases} u_{i,j+1} - u_{i,j} & \text{if } j < N \\ 0 & \text{if } j = N. \end{cases}$$

We define a new vector-valued image by composing these gradients as follows: when $m_{ij} = 0$ (non-face pixel) we use the gradient of u_2, and when $m_{ij} = 1$ (face pixel) we use the gradient of u_1. We store this in a new vector valued image $v : \Omega \to \mathbb{R}^2$:

$$\begin{aligned} v_{ij} &= m_{ij} \nabla^+ u_{1,ij} + (1 - m_{ij}) \nabla^+ u_{2,ij} \\ &= \begin{bmatrix} m_{ij} \nabla_i^+ u_{1,ij} + (1 - m_{ij}) \nabla_i^+ u_{2,ij} \\ m_{ij} \nabla_j^+ u_{1,ij} + (1 - m_{ij}) \nabla_j^+ u_{2,ij} \end{bmatrix} \\ &= \begin{bmatrix} v_{1,ij} \end{bmatrix} \quad \blacklozenge \end{aligned}$$

$$\lfloor v_{2,ij} \rfloor$$

for $(i,j) \in \Omega$.

Now we have to find an image u whose gradient coincides (or is similar) with v: $\nabla^+ u \approx v$. In addition, the image should be u_2 outside the face (when $m_{ij} = 0$). We will find u by solving a quadratic energy which imposes these too conditions as follows:

$$E(u) = \overbrace{\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{M} |\nabla^+ u_{ij} - v_{ij}|^2_{\mathbb{R}^2}}^{\text{gradients similar to } v} + \overbrace{\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{M} \beta_{ij}(u_{ij} - u_{2,ij})^2}^{u \text{ is } u_2 \text{ outside the mask}}$$

where $\beta : \Omega \to \mathbb{R}$ is a coefficients image which controls the attachment to $u_{2,ij}$. We will use $\beta_{ij} = \beta_0(1 - m_{ij})$, where $\beta_0 \in \mathbb{R}$ is a constant. This means that β_{ij} is zero on the face region, only acting outside the face. Recall that $|\cdot|$ denotes the Euclidean norm in $\mathbb{R}^2$.

Similar to what we did for denoising, we can express the energy in matrix notation, using our vector representation of images and discrete gradients:

$$E(u) = \frac{1}{2}\langle \nabla^+ u - v, \nabla^+ u - v\rangle_{\mathcal{Y}} + \frac{1}{2}\langle B(u - u_2), u - u_2\rangle_{\mathcal{X}}.$$

where B is a $MN \times MN$ diagonal matrix, which in its diagonal holds the image β_{ij}. The rest of the notation is as in the previous assignment, $\mathcal{X} = \mathbb{R}^{MN}$ denotes the space of gray-scale images, and $\mathcal{Y} = \mathbb{R}^{2MN}$ denotes the space of vector-valued images. $\langle \cdot, \cdot \rangle_{\mathcal{X}}$ is the scalar product in $\mathbb{R}^{NM}$ and $\langle \cdot, \cdot \rangle_{\mathcal{Y}}$ is the scalar product in $\mathbb{R}^{2NM}$.

Recasting as a quadratic optimization problem

We have expressed our editing problem as a quadratic energy, however we still need a bit of work to cast the previous expressions into a more familiar way, namely,

$$E(u) = \frac{1}{2}\langle u, \mathbf{A}u\rangle - \langle b, u\rangle + c,$$

where $c \in \mathbb{R}$ is a constant, $b \in \mathcal{X}$ and $\mathbf{A}$ is a $MN \times MN$, symmetric, positive definite matrix.

To do so, let us define the backward divergence operator div^- (based on backward differences) for a vector image

$$u = \begin{bmatrix} u^{(1)} \\ u^{(2)} \end{bmatrix}$$

as follows:

$$(\mathrm{div}^- u)_{i,j} = \begin{cases} u^{(1)}_{i,j} - u^{(1)}_{i-1,j} & \text{if } 1 < i < M \\ u^{(1)}_{i,j} & \text{if } i = 1 \\ -u^{(1)}_{i-1,j} & \text{if } i = M \end{cases} + \begin{cases} u^{(2)}_{i,j} - u^{(2)}_{i,j-1} & \text{if } 1 < j < N \\ u^{(2)}_{i,j} & \text{if } j = 1 \\ -u^{(2)}_{i,j-1} & \text{if } j = N \end{cases}$$

One can show that ∇^+ and $-\mathrm{div}^-$ are **dual** to each other, meaning that for any u and v of appropriate dimensions, it holds that

$$\langle \nabla^+ u, v\rangle_{\mathcal{X}} = \langle u, -\mathrm{div}^- v\rangle_{\mathcal{Y}}.$$

Using this duality, we can express the L^2 norm of a forward gradient as follows:

$$\|\nabla^+ u\| = \langle \nabla^+ u, \nabla^+ u\rangle_{\mathcal{Y}} = \langle -\mathrm{div}^- \nabla^+ u, u\rangle_{\mathcal{X}}.$$

This allows us to manipulate the terms of the energy E, as follows:

$$\begin{aligned} E(u) =& \frac{1}{2}\langle \nabla^+ u - v, \nabla^+ u - v\rangle_{\mathcal{Y}} + \frac{1}{2}\langle B(u - u_2), u - u_2\rangle_{\mathcal{X}} \\ =& \frac{1}{2}\langle \underbrace{(B - \mathrm{div}^- \nabla^+)}_{\mathbf{A}} u, u\rangle_{\mathcal{X}} - \langle \underbrace{Bu_2 - \mathrm{div}^- v}_{\mathbf{b}}, u\rangle_{\mathcal{X}} + \underbrace{\frac{1}{2}\langle v, v\rangle_{\mathcal{Y}} + \frac{1}{2}\langle Bu_2, u_2\rangle_{\mathcal{X}}}_{c} \end{aligned}$$

Assignment

1. Read the paper on Poisson editing
2. Prove the duality relation between forward gradient and backward divergence
3. Implement the Poisson editing based on the code provided
4. Apply your Poisson editing on different pictures of your own choice, try out variant from the original paper, and comment